

KITEWRIGHT CORE · PRECIS

Kitewright Core design precis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	TRL 2 and TRL 3 precis on the constructable design; decisions of KWC-DDR-001 and KWC-DDR-002 under Amish's 2026-10-03 pre-approval; numbers from KWC-CAL-001
0.1	2026-09-30	Amish Chadha	Initial scaffold

Kitewright Core design precis

The shared core of the Kitewright open drone family: open autopilot, ColdCell power bus and one standard payload mount.

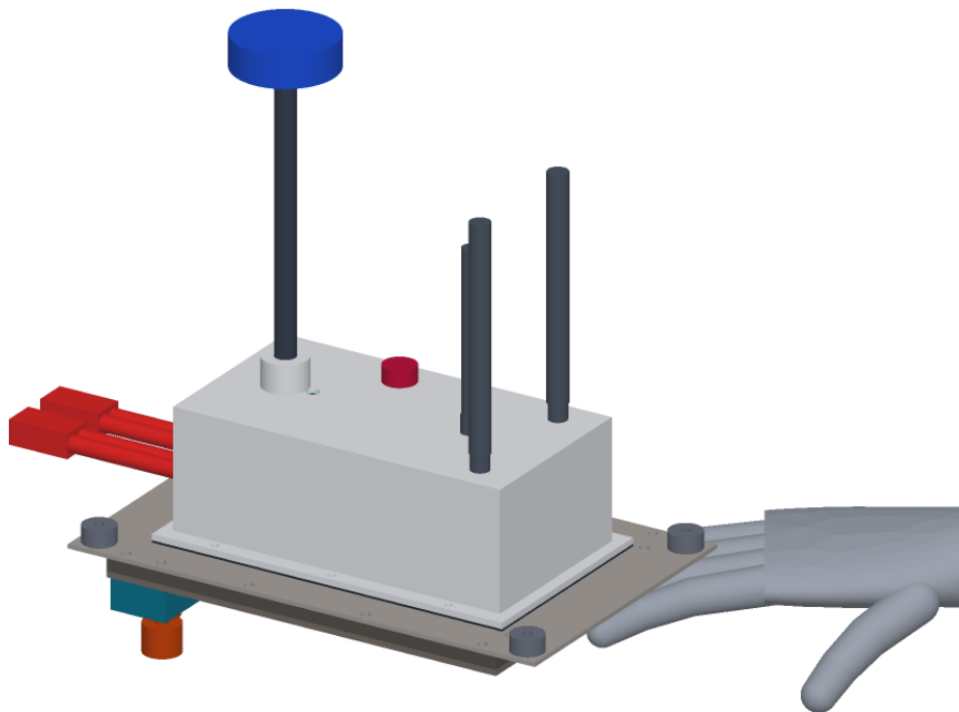
■ Summary

Kitewright Core is a 240 x 150 mm aluminium plate that hangs under a frame's deck on four M4 bolts. On top, under a printed lid that rises through an opening in the deck, sit a Pixhawk-standard flight controller on vibration dampers, a power distribution board for two ColdCell packs, two independent 5.3 V supplies, a payload power switch, a telemetry radio and a control link receiver. A GNSS receiver stands on a removable mast above the lid. Under the plate is the payload mount: a plain rail of aluminium bars in which a 5 mm payload shoe slides from the rear to two front stops and is held by two spring locking pins, with the DS-014 power and data plug on a short pigtail. The same core, unchanged, flies Kitewright Lift and Kitewright Range with published parameter sets for three altitude bands. On paper it meets ten of its eleven requirements; its estimated mass, 1.27 kg, misses R9 (Figure 1, KWC-CAL-001).

Kitewright Core

github.com/BoujeeEnjinia1701/kitewright-core

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey: Adult hand (scale) for scale

Figure 1. Kitewright Core, generated from the parametric model. CONCEPT, NOT FOR FABRICATION.

■ How it works

ColdCell packs plug into the two AS150 inputs at the back of the lid. The power distribution board joins them, measures total current with a Hall sensor and feeds the frame's speed controllers through two AS150 outputs. Two independent supplies, a digital power monitor and a backup supply, each turn the 18 to 60 V bus into 5.3 V for the flight controller, so one failure does not drop the avionics. A payload switch, enabled by the flight controller, feeds the bus to the payload's battery pins through the DS-014 pigtail with an 8 A limit.

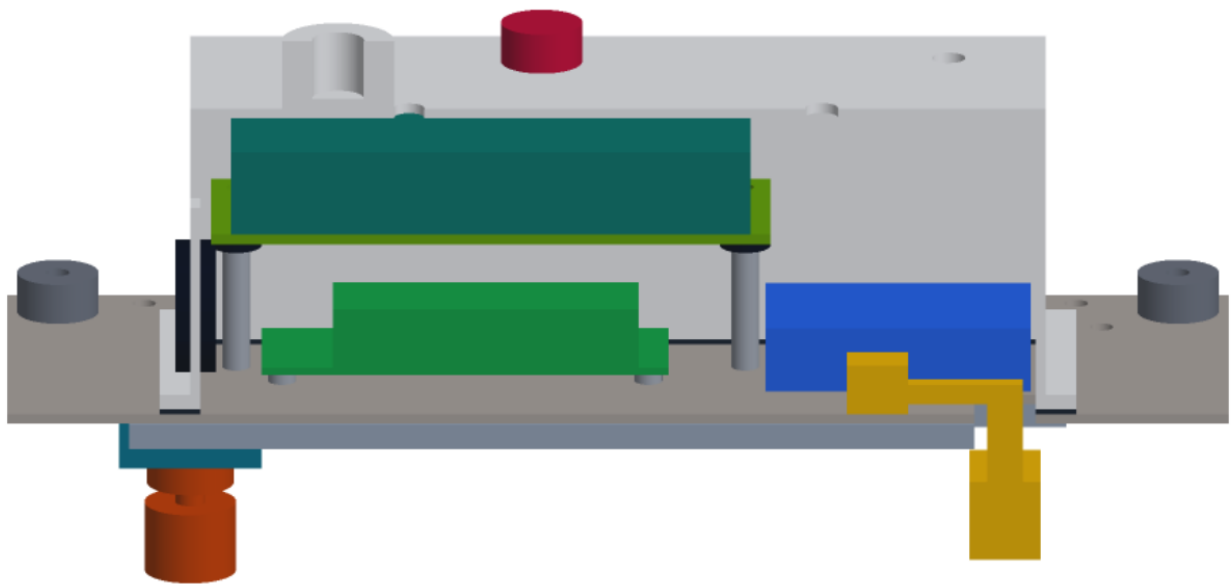
The flight controller runs released PX4 with Kitewright parameter sets. It reads the packs' temperature and charge over DroneCAN from ColdCell's battery management system, the GNSS and compass on the mast, and the barometer on its own board, and it talks to the ground station over a 900 MHz class telemetry radio and to the pilot's handset over a long-range control link.

A payload carries a 184 x 128 x 5 mm aluminium shoe on its top. The operator pulls and twists both locking pins to their rest, slides the shoe into the rail from the rear until it stops against the front stops, and twists the pins back: they snap up into the shoe's two holes. The DS-014 plug is then pushed into the payload's socket, and the flight controller recognises the payload over MAVLink. The shoe's edges run between the plate and the taped lips with a 0.3 mm gap, so the lips carry its weight; the pins and front stops hold it fore and aft.

Kitewright Core: cutaway

github.com/BoujeeEnjinial701/kitewright-core

CONCEPT, NOT FOR FABRICATION



Front half removed; seen from the front and above, 18 deg elevation

Figure 2. Cutaway, front half removed: the damped flight controller above the power board, and the rail and shoe below the plate.

Kitewright Core: power flow in Lift hover at sea level, W (KWC-CAL-001 estimates)

github.com/BoujeeEnjinia1701/kitewright-core

CONCEPT, NOT FOR FABRICATION

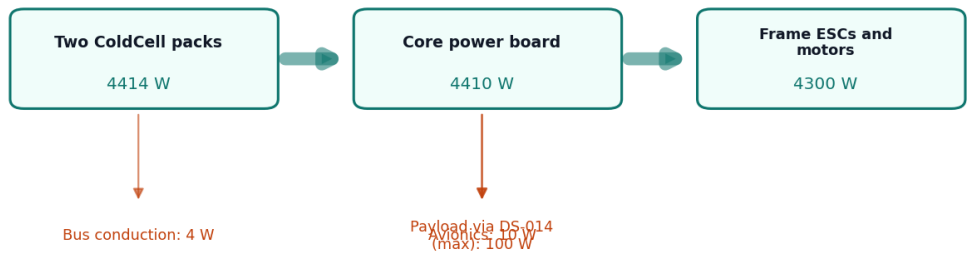


Figure 3. Power flow in Kitewright Lift's hover at sea level (KWC-CAL-001 estimates).

■ Main components

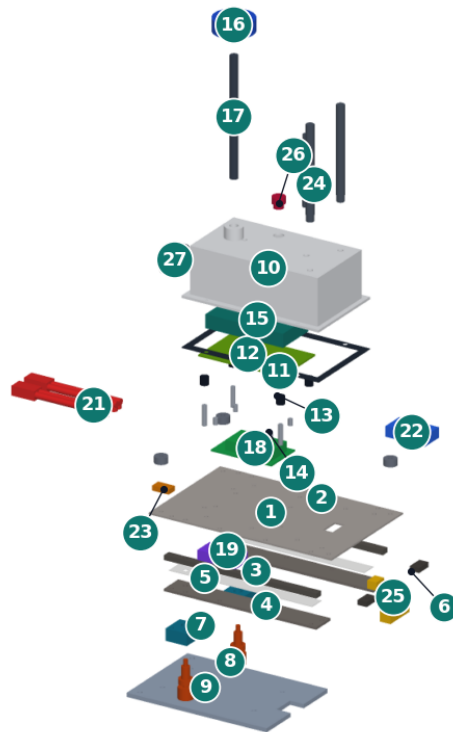
Table 1. Components, numbered as in `bom/bom.csv` and Figure 4.

#	Component	Role
1	Core plate, 240 x 150 x 2 mm 5052 aluminium	Carries everything; bolts to the frame on a 220 x 130 mm pattern
2	Corner spacers (4), 16 x 8 mm	Hold the plate 8 mm under the frame deck
3, 4, 5	Rail spacer bars, lips and wear tape	The plain rail: 200 mm long, 19 mm lip overlap, 0.3 mm running gap
6	Front stops (2)	End of travel, where both pins line up with their holes
7, 8	Pin blocks and locking pins (2)	Two independent spring plungers; a red band shows a withdrawn pin
9	Payload shoe	Fitted to every payload; the core kit carries one
10, 11	Lid and gasket	Printed ASA cover with mast boss, antenna, switch and grommet holes
12, 13, 14	Damping plate, dampers, standoffs	Isolate the flight controller from vibration
15	Flight controller, FMUv6X class	Runs PX4; triple IMU with heater; 16 outputs; Ethernet and CAN
16, 17	GNSS receiver and compass, mast	Position and heading, 231 mm above the plate, clear of power wiring
18, 19	Power distribution board, power modules	Pack inputs, frame outputs, current sensing, two 5.3 V supplies, payload switch
20	Clinch nuts (14)	Flush threads under the plate where the shoe slides
21	Power leads and AS150 connectors	Two pack inputs and two frame outputs, 8 AWG
22, 23, 24	Telemetry radio, control link receiver, antennas	Command, telemetry and control path
25	DS-014 connector pair and pigtail	Power and data to the payload
26 to 29	Safety switch, grommets, harness, fasteners	Arming safety, cable entries, cold-rated wiring
30	Companion computer (optional)	Payload processing in its own box on the frame, over Ethernet
31 to 34	Handset, ground station tablet, ground antenna, field charger	Ground station kit
35	Spares kit	Field spares

Kitewright Core: exploded viewgithub.com/BoujeeEnjinia1701/kitewright-core

CONCEPT, NOT FOR FABRICATION

- 1 Core plate
- 2 Corner spacers (4)
- 3 Rail spacer bars (2)
- 4 Rail lips (2)
- 5 Wear tape
- 6 Front stops (2)
- 7 Pin blocks (2)
- 8 Locking pins (2)
- 9 Payload shoe
- 10 Lid
- 11 Lid gasket
- 12 Damping plate
- 13 Vibration dampers (4)
- 14 Standoffs (8)
- 15 Flight controller
- 16 GNSS receiver and compass
- 17 GNSS mast
- 18 Power distribution board
- 19 Power modules
- 21 Power leads and AS150 plugs
- 22 Telemetry radio
- 23 Control link receiver
- 24 Antennas
- 25 DS-014 pigtail and plug
- 26 Safety switch
- 27 Grommets (2)



Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

Figure 4. Exploded view; numbers match `bom/bom.csv`.

■ Numbers from the TRL 3 calculations

Table 2. Key figures (KWC-CAL-001).

Quantity	Value	Basis
Rated payload	5 kg; vertical design load 147 N	R3
Lowest strength factor	10.8 (plate edge as a beam between frame bolts)	CAL section A
Fore-aft load on one pin alone	221 N; 11.2 MPa shear in a 5 mm pin	CAL section A
Payload swap with gloves	About 52 s, no tools	CAL section B
Payload current for 100 W	5.6 A at 18 V (6S), 2.2 A at 44.8 V	CAL section C
Hottest part at 45 °C ambient	About 75 °C (parts rated 85 °C)	CAL section D
Air density at 5,000 m	0.736 kg/m ³ , 60 % of sea level	CAL section E
Hover time at 5,000 m and -20 °C	47 % of sea level with cold packs; 70 % with ColdCell	CAL section F
Flight controller outputs used	Lift 11, Range 10, of 16	CAL section G
Core mass	About 1.27 kg (R9 not met)	CAL section H
Cost	USD 3,940.60 with ground station and spares	CAL section I

Value-engineering target: USD 5,000. Estimated cost of the constructable design: USD 3,940.60 (USD 1,059.40 under the target).

■ Key design choices

All decided under Amish's 2026-10-03 pre-approval; each is argued in KWC-DDR-001 or KWC-DDR-002 and indexed in the design decisions register, KWC-DEC-001.

- **PX4 first, ArduPilot as the documented alternative** (D1), on a Pixhawk FMUv6X class controller (D2).
- **One bus, 18 to 60 V** (D3), so a 6S Range and a 12S or 16S Lift use the same board, supplies and switch.
- **One payload class for version 1: 5 kg and 100 W** (D4).
- **Plain rail, rear entry, front stops, pigtail plug** (D5); no bayonet and no blind-mating connector.
- **Two independent locking pins** (D6), the conservative safety choice; either holds the payload alone.
- **Core hangs under the frame deck** (D7) on a 220 x 130 mm M4 pattern; the lid rises through a 200 x 112 mm opening.
- **Companion computer outside the core** (D8), so its heat and mass stay with the frames that need it.
- **ColdCell reports over DroneCAN** (D9), which released PX4 reads without changes.

■ Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice), all kept in the constructable design:

- Payload mount uses the Pixhawk DS-014 connector with a plain rail and locking pins, no bayonet, to stay clear of DJI US 2021/0316882 (payload mounting platform, claims unread). DJI US9890900B2 (quick gimbal connector) is expired.
- ColdCell packs use an external film heater on standard cells with a BMS thermostat, not in-cell resistor sheets (EC Power US9882197B2, to 2036). The core only reads the packs.

- Any tether input to the power bus is a fixed-voltage supply with onboard DC-DC conversion, not adaptive tether voltage (Elistair US11059580B2, to 2037). The core's 18 to 60 V inputs take a fixed-voltage tether converter like a pack.
- Attorney claim reads before any sale: DJI 2021/0316882, US9882197, US11059580.

■ Relationship to other lab projects

- ColdCell (heated battery packs; reports over DroneCAN on the core's bus)
- AltiRig (thin-air propeller and motor test chamber; supplies the hover-thrust data for Table 6 of KWC-CAL-001)
- Kitewright Lift and Kitewright Range (host frames; provide the deck pattern and opening of D7)
- AvalancheScout and LakeWatch (payloads; each carries a shoe of KWC-DWG-106 and a DS-014 socket)

■ Safety

Safety: Published as an open engineering reference, not certified aviation equipment. Civilian use only. No weapons, targeting or military payloads, and none will be accepted into the family.

Safety: Certification and operating approvals (FAA Part 107 in the United States, EASA rules in Europe, India's DGCA Drone Rules 2021) are out of scope at this TRL and will be addressed if prototypes progress. Builders must fly only where local rules allow.

Safety: Lithium packs: the core carries up to 60 V and 200 A peak from two packs. Connect packs only through the anti-spark AS150 connectors, check polarity on a current-limited bench supply first, never charge a pack below its charge temperature (ColdCell blocks this), and keep a lithium fire plan at hand. A crashed pack can catch fire hours later.

Safety: Moving machinery: propellers on the frames can cause serious injury. The core's safety switch must be pressed before arming; arm only in a clear area, and write pre-flight, arming and keep-out procedures before any powered test.

Safety: Falling payload: a released payload is a falling object. Two independent locking pins are fitted; both must show no red band, and the payload is tugged, before every flight. Nobody stands under a payload being lifted.

Safety: Loss of link or power in thin air can mean a crash in remote terrain: return-to-launch, low-battery and geofence failsafes must be set and tested at each altitude band (KWC-CAL-001, Table 6).

This design is published as an open engineering reference. It is not certified equipment.

■ Open questions

None for design. Facts that can only be settled with parts in hand (the DS-014 licence, pinout and pin rating, the controller's mounting pattern, plunger stroke) are listed under "To confirm when parts are bought" in KWC-DEC-001. R9 core mass awaits Amish's decision (docs/REVIEW.md).